



WA-7438

Integrated M. C. A. (Sem. IV)
Examination, June - 2024

IMCA - 403

Computer Oriented Numerical Methods

Time Allowed : Three Hours

Max. Marks : 70

Min. Pass Marks : 28

Note : This question paper contains three sections as under :

Section-A**Max. Marks-10**

This section shall have 01 compulsory question comprising 10 questions (maximum 20 words answer) taking two questions from each unit. Each question shall be of one mark and total marks of this section will be 10. This section will be compulsory in the paper.

Section-B**Max. Marks-25**

This section will carry 25 marks with equally divided into five long answer type questions (answer about in 250 words) and students are instructed to attempt five questions by selecting one question from each unit.

Section-C**Max. Marks-35**

Will carry 35 marks with five long answer type questions comprising one compulsory question (question no. 07) of 15 marks and four questions of 10 marks each. Students are instructed to attempt total three questions with one compulsory question (answer about in 500 words) and any two more questions (answer about in 400 words) out of remaining four questions.

SECTION - A**1**

- (i) Binary Equivalent of 62.31 is _____.
- (ii) Error in R-K method is of order _____.
- (iii) Find $\sqrt{7}$ using Newton Raphson method.
- (iv) State the Gauss two point formulae.
- (v) What is the condition for a system of linear equation to be solvable by Gauss elimination or Jacobi method?
- (vi) What is Boundary value problem?
- (vii) Find 101101 – 100111 in Binary.
- (viii) Write following system of linear equation in matrix form :

$$-8x + 6y = 13$$

$$9x - 5y = 7$$

- (ix) Write the formula used in Regula Falsi method for finding roots of an equation.
- (x) Construct a difference table for following data :

x	1	2	3	4
f(x)	2	9	28	65

$$1 \times 10 = 10$$

SECTION - B

UNIT - I

- 2 Obtain the smallest positive root of the equation $x^3 - 5x + 1 = 0$ by using three iterations of bisection method.

OR

Describe relative merits of each of direct methods and iterative methods of solving system of linear equations, over each other.

UNIT - II

- 3 Solve $x^3 - 5x + 3 = 0$ Regula-Falsi method.
OR

Use bisection method to approximate the value of $\sqrt{3}$ given that it lies in interval $[1, 2]$.

UNIT - III

- 4 Solve the following system of linear simultaneous equations by Gauss elimination method :

$$2x + 2y + z = 12$$

$$3x + 2y + 2z = 18$$

$$5x + 10y - 8z = 10$$

OR

Solve following system of linear simultaneous equations by Gauss Jordan method :

$$2x + y + 6z = 9$$

$$8x + 3y + 2z = 13$$

$$x + 5y + z = 7$$

UNIT - IV

- 5 Find Newton's Backward difference form of interpolating polynomial for the following data :

x	3	5	7	9	11	13
$f(x)$	16	36	64	100	144	196

Hence evaluate $f(12)$.

OR

Solve the following Boundary value problem

$$\frac{d^2 y}{dx^2} - y = 0 \text{ with } y(0) = 0.$$

$$y(2) = 3.62686$$

UNIT - V

- 6 Write flow chart to solve the following $\frac{dy}{dx} = f(x, y)$ with initial condition

$$y(x_0) = y_0 \text{ by Euler's method.}$$

OR

Write short notes on :

- (1) Curve fitting
- (2) Trapezoidal Rule

SECTION - C

- 7 (a) Solve the system using Gauss-Elimination method :

$$3x + y - z = 3$$

$$2x - 8y + z = -5$$

$$x - 2y + 9z = 8$$

- (b) Find Eigen value, Eigen vector of the matrix

$$\begin{bmatrix} -1 & -7 & -5 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{bmatrix}$$

- 8 Solve the system of equations

$$8x - y + z = 18$$

$$2x + 5y - 2z = 3$$

$$x + y - 3z = -6$$

by using Gauss Seidel iterative method.

- 9 Compute the value of $\int_4^{5.2} \log_e x \, dx$ by using Simpson's $\frac{1}{3}$ rule. (take $h = 0.2$)

- 10 From the following data find y at $x = 43$ by using Newton's forward interpolation formula.

x	40	50	60	70	80	90
$f(x)$	184	204	226	250	276	304

- 11 Use Runge-Kutta 4th order method with step size 0.5 to solve the initial value problem.

$$y' = \frac{3x}{y} - xy, y(0) = 2 \text{ over the interval } [0, 1].$$